

mature. If there is something better, we move to it. New processes won't be ready any time soon."

Another alternative wafer production method is to etch the LED arrays for bonding to the circuit. While this method has less accuracy problems, it is very slow and also needs improvement to etch miniature LED components.

Fluid assembly technique, interpose, and self-assembly transfer are some of the trial and error methods that are widely practiced. However, the improvements to yield the desired level of transfer still needs exploration.

Not only are the initial stages of production difficult, the following stages of bonding add to the difficulty. The  $\mu$ LED chips are so small in size that the soldering paste may cause a short circuit. Inspection and testing stages also pose challenges due to the large number of chips, and hence manufacturers have to look for mass test methods to address it accurately. If a defective product is identified, the process of repairing it is much more complex. Currently, the  $\mu$ LED recovery solutions use UV irradiation rescue, laser rescue, and so on.

## Supply chain and R&D focus

Currently, companies are relying on existing advantages of having complete mature supply chains for LED and display panels. The supply chains of  $\mu$ LEDs are a result of cross industry collaboration between electronic brands and leading edge technology players.

Apart from LED manufacturers, mini and  $\mu$ LED developers take help of mass transfer technology startups, semiconductor chip makers, panel makers, and production equipment suppliers. Display panel manufacturers, who are shifting their focus to mini and  $\mu$ LEDs, have also led more developers to step up R&D for new products.

In addition to panel manufactur-

## News worth a look

**A way to reduce incidental expenses in production.** A Korean research team from the Electronics and Telecommunications Research Institute (ETRI) has announced it has found methods to simplify the process and reduce the incidental expenses of  $\mu$ LED production. While all the existing methods are based on sequential transfer, bonding process, the team claims to have developed new materials, simultaneous transfer, and bonding (SITRAB) adhesives for use in combining transfer and bonding into one process.

**GaN based RGB  $\mu$ LED displays.** Porotech, a startup in the UK, focuses on developing wide-bandgap compound GaN  $\mu$ LED devices built with etching to create porous GaN. Using porous technology, Porotech has shown InGaN red emissive  $\mu$ LEDs that can be used in applications such as AR and VR. It also claims it can make green and blue  $\mu$ LEDs on the same substrate. (Source: [www.porotech.co.uk](http://www.porotech.co.uk))

**$\mu$ LED based stretchable display.** Royole Corporation, a supplier of flexible electronics, claims to have developed the world's first  $\mu$ LED based stretchable display technology. It demonstrated a 6.86cm (2.7-inch) 96 $\times$ 60 resolution stretchable panel offering 130% stretchability and up to 40-degree convex bending. The  $\mu$ LED stretchable display technology is also said to reach a resolution of up to 120 pixels per inch (PPI) and allow much higher transmittance than current flexible OLED display technologies. The high transmittance of up to 70% makes Royole's solution highly applicable for smart car windshields and sunglasses. (Source: <https://global.royole.com>)

**Laser integrated system for  $\mu$ LED manufacturing.** Coherent, a photonic solutions provider, has announced a new UVtransfer integrated laser system that performs the three vital processes in  $\mu$ LED fabrication—laser lift-off (LLO), laser-induced forward transfer (LIFT), and repair/trimming. It's based on excimer laser technology and covers areas as large as 16 $\times$ 2 mm<sup>2</sup> per shot, enabling massive parallel device processing. This is critical in moving hundreds of millions of dies for large displays and making these displays economically viable. (Source: [www.coherent.com](http://www.coherent.com))

ers, LED producers have practically made it a goal to commercialise  $\mu$ LED and mini LED. The industry has noted an increase in the supply chain, especially in Taiwan and China where manufacturers like Epistar, PlayNitride, HC Semitek, and many others are working on developing mini and  $\mu$ LED technology. "There are definitely lots of technological challenges, and supply chain incompleteness. Therefore, targeting correct applications as the entry market is important," suggests Dr Xiaoxi.

## Inevitable price war

Liu tells that  $\mu$ LED is currently in the phase of exploring new opportunities, transparent displays, indoor ultra-large displays (like 100 or 200 inches), and applications in automobiles. "The cost would be immense to compete with existing display technologies like LCD or OLED for now," he adds.

The Taiwan government offers large subsidies to its domestic

companies, encouraging mini and  $\mu$ LED development. Liu shares that even manufacturing associations have teamed up for innovations and funds to upscale production. "Government can support in promoting technology and import tax issues," he adds.

Whatsoever, there is a concern whether the increase in capacity will lead to oversupply and price war. Industry sources opine that a large number of competitors may not be that bad. Such an increase in competitors is likely to expand the scale of the market, which in turn may lead to a breakthrough in innovations. Striking a balance between price and specifications becomes a necessity to meet affordability expectations of the market. The price war, however, could be inevitable for companies that focus on aggressive price strategy to stimulate and control demand.

Instead of worrying about the upcoming price war, mini and  $\mu$ LED developers should address techno-